① $V_1 = k_1[S_1]$　　$V_2 = k_2[S_2]$

$$\frac{d[S_2]}{dt} = V_1 - V_2$$

$$= k_1[S_1] - k_2[S_2]$$

② (1) $$\frac{d[Glucose]}{dt} = -k[Glucose]$$

$$\int \frac{d[Glucose]}{[Glucose]} = \int -k\,dt$$

$$\log[Glc] = -tkt + C \quad (C\text{は積分定数})$$

$$\log[Glc] = -tk + C$$

$$[Glc] = e^{-tk + C}$$

$$[Glc] = C' e^{-tk}$$

$t = 0$ のとき

$$[Glc](t=0) = 1.0\,mM = C' \times e^{-0k}$$

$$= C' \times 1.0$$

つまり、$C' = 1.0$ より

$$[Glc] = e^{-tk}$$

③ matplotlib で書く

(4) (1) $$\frac{dS_1}{dt} = -\frac{V_{m1}[S_1]}{K_{m1}+[S_1]}$$

$$\frac{dS_2}{dt} = \frac{V_{m1}[S_1]}{K_{m1}+[S_1]} - \frac{V_{m2}[S_2]}{K_{m2}+[S_2]}$$